ISRO Scientist or Engineer Electronics 2009

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Electronics

Instructions
For the following questions answer them individually

Question 1

If a counter having 10 FFsis initially at 0, what countwill it hold after 2060 pulses?

A 000 000 1100

B 000 001 1100

C 000 001 1000

D 000 000 1110

Answer: E

Question 2

Amemory system of size 16 K bytes is required to be designed using memory chips which have 12 address lines and 4 datalines each. Then the numberofsuch chips required to design the memory system is

A 3

B 4

C 8

D 16

Answer: E

Question 3

Consider the following program for 8085
XRAA
LXI B, 0007H
LOOP: DCXB
JNZ LOOP
The loop will be executed

A 8 times

B once

C 7 times

D infinite times

Answer: E

Question 4

Acertain JK FF has t_{pd}=12 ns. The largest MOD counter that can be constructed from such FFs and still operate up to 10 MHz is

A 16

B 256

C 8

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D 128

Answer: E

Question 5

Convolution of x(t+5) with impulse function δ (t-7) is equal to

A x(t - 2)

B x(t + 12)

C x(t - 12)

D x(t + 2)

Answer: E

Question 6

The purpose of a Cassegrain feed in a parabolic reflector antenna is to

A Achieve higher antenna gain

B Reduce the antenna size

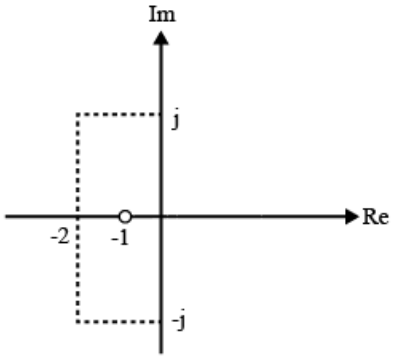
C Reduce the beamwidth

D Ease of locating the feed at convenient point

Answer: E

Question 7

A transfer function G(s) has pole-zero plot as Im "shown in the figure. Given that the steady state function gain is 2,, the transfer function A G(s) will be given by



A $\frac{5(s+1)}{s^2+4s+4}$

B $\frac{2(s+1)}{s^2+4s+5}$

C $\frac{10(s+1)}{s^2+4s+5}$

D $\frac{10(s+1)}{(s+2)^2}$

Answer: E

Question 8

The impedance of a series RLC circuit is

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A $\sqrt{R^2 + X_l^2 + X_c^2}$

B $\sqrt{R^2 + X_l^2 - X_c^2}$

C $\sqrt{R^2 + (X_l + X_c)^2}$

D $\sqrt{R^2 + (X_l - X_c)^2}$

Answer: E

Question 9

The advantage of $\frac{\pi}{4}$ QPSK compared to conventional QPSKis ?

A $\frac{\pi}{4}$ QPSK requires less bandwidth compared to conventional QPSK

B $\frac{\pi}{4}$ QPSKproduces lower envelope variation compared to conventional QPSK

C $\frac{\pi}{4}$ QPSKreceiverdesign is simple compared to» conventional QPSK

D There is no advantageA movingcoil iron ammeter may be compensated for frequency errors by

Answer: E

Question 10

A movingcoil iron ammeter may be compensated for frequency errors by

A Series inductance

B Shuntresistance

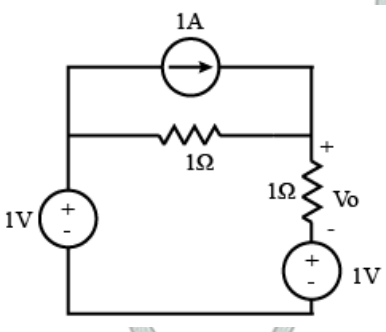
C Series resistor.

D Shunt capacitor

Answer: E

Question 11

Find the value of V_o in the figure.



A $\frac{1}{2}$ V

B $\frac{-1}{2}$ V

C 0 V

D $\frac{-3}{2}$ V

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Answer: E

Question 12

What is the approximate skin depth in copper at 100 MHz?

A 0.1 micron

B 10 microns

C 10 mm

D 100 mm

Answer: E

Question 13

While designing a low noise amplifier, what is the importanceof the noise resistance of a transistor?

A It gives the effective impedance offered by the input with respectto noise current

B It defines the criterion for conjugate matching of input impedance

C It tells us the resistance that would generate the same amountof noise at room temperature

D It tells us how rapidly the noise figure increases as we move away from the optimum source impedance

Answer: E

Question 14

Unattenuated radiation field at the surface of the earth of a quarter-wave monopole will exist if the earth surfaceis

A Lossy dielectric

B Perfectly conducting

C Perfect insulator

D None of these

Answer: E

Question 15

Given that $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

Evaluate $A^3 - 6A^2 + 11A - 10I$

A Null matrix

B Identity matrix

C -4I

D None of the above

Answer: E

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Question 16

$\lim_{x \to 0} x \log \sin(x)$ **is**

A 0

B $\frac{1}{2}$

C 1

D 2

Answer: E

Question 17

A12 bit ADC is operating with a 1 ps clock period and the total conversion time is seen to be 14 μ s. The ADC must be of

A Flash type

B Counting type

C Integrating type

D Successive Approximation type

Answer: E

Question 18

Which of the following types of devices is not field programmable?

A FPGA

B ASIC

C CPLD

D PLD

Answer: E

Question 19

Which is the correct orderofdifferent process steps for a typical FPGA design?

A Functional simulation, Synthesis, Place & Route, Timing Verification

B Functional simulation, Timing Verification, Synthesis, Place & Route

C Timing Verification, Synthesis, Functional simulation, Place & Route

D Synthesis, Functional simulation, Timing Verification, Place & Route

Answer: E

Question 20

The theoretical dividing line between Reduced Instruction Set computing (RISC) microprocessor and Complex Instructions Set Computing (CISC) microprocessoris

A Instruction execution rate to be one instruction per clock cycle

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B Numberof address and data lines

C Numberofpinsin the chip

D None of the above

Answer: E

Question 21

Ina pulsed radar, the peak radar transmitter poweris increased by a factor of 81 keeping all other radar parameters unchanged. The maximum range capability of the radar increases by a factor of

A 81

B 9

C 27

D 3

Answer: E

Question 22

A radar transmitter has a PRF of 1500 PPSand a pulse width of 1 micro second. Its duty cycle is

A 0.0075

B 0.015

C 0.6015

D 6.00075

Answer: E

Question 23

Whichofthe following technology results in least power dissipation 2

A CMOS

B ECL

C TTL

D NMOS

Answer: E

Question 24

A power MOSFET is a

A Current controlled device

B Frequency controlled device

C Voltage controlled device

D None of the above

Answer: E

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Question 25

The advantage of differential PSK (DPSK) over coherent PSKis

A DPSK requires less bandwidth compared coherent PSK

B DPSKreceiver design is simple compared to coherent PSK

C DPSK Bit Error rate is lower than coherent PSK

D For same bandwidth, DPSKbit rate is higher compared to coherent PSK

Answer: E

Question 26

An ideal current meter should have

A Zero resistance

B Finite resistance

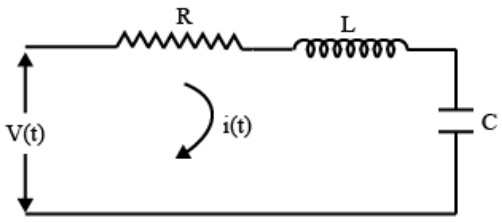
C Infinite resistance

D Very large resistance

Answer: E

Question 27

For the network shown in the figure V;(t) is the input and i(t) is the output. The transfer function $\frac{I(s)}{v_i(s)}$ of the network is



A $\frac{C}{RCs^2+LCs+1}$

B $\frac{C}{LCs^2+RCs+1}$

C $\frac{Cs}{RCs^2+LCs+1}$

D $\frac{Cs}{LCs^2+RCs+1}$

Answer: E

Question 28

The length of a lossy shorted stub is $\frac{1}{32}\lambda$ he Whatwill be the natureof its input impedance?

A A low inductive reactance in series with a low resistance

B A low capacitive reactance in shunt with a high resistance

C A high capacitive reactance in series with a low resistance

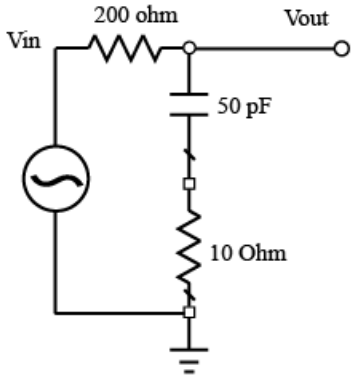
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D A high inductive reactance in series with a low resistance

Answer: E

Question 29

Refer to the circuit in the figure. Which of the following statements is NOT true about this circuit?



A It is a low passfilter

B It has a finite attenuation at infinite frequency

C Its phase shift varies monotonically with the frequency

D It has one pole and one zero

Answer: E

Question 30

Which of the followingis the correct continuity equation:

A $\nabla \bullet J = \frac{\partial \rho}{\partial t}$

B $\nabla \bullet J = -\frac{\partial \rho}{\partial t}$

C $\nabla \bullet J = 0$

D $\nabla \bullet J = -\frac{\partial E}{\partial t}$

Answer: E

Question 31

α, β are the roots of the equation $x^2 + x + 1 = 0$. Then $\alpha^{3n} + \beta^{3n}$ is

A 1

B 2

C 0

D 3

Answer: E

Question 32

A point moving in the complex plane satisfies the following relation $z^2 + z^{*^2} = 8$, where z^* stands for the complex conjugate of z. The difference of the distances of the moving point from $(2\sqrt{2}, 0)$ and $(-2\sqrt{2}, 0)$ is

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A 8

B 2

C 4

D 6

Answer: E

Question 33

The greatest negative number which can be stored ina computer that has 8-bit word length and uses 2's complementarithmeticis

A -256

B 255

C -128

D -127

Answer: E

Question 34

Which of the following ADCs uses over sampling in its operation

A Sigma — delta ADC

B Counter ramp convertor

C Successive Approximation Register ADC

D Flash Convertor

Answer: E

Question 35

A mealy state machine's output depends on

A State and outputs

B Inputs

C State

D State and inputs

Answer: E

Question 36

The characteristic equation of the $T-FF$ is given by

A $Q^{+}=TQ$

B $Q^{+}=T^{Q}$

C $Q^{+}\ =TQ$

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D $Q^{+} = TQ + QT$

Answer: E

Question 37

Percentage modulation of an AM wave having a powercontent of 8 KW at carrier frequency and 2 KW in each of its side bandsis

A 60 %

B 70 %

C 100 %

D 80 %

Answer: E

Question 38

Approximate equivalent noise temperature (deg. K) of an amplifier with a noise factor of 1.04 is

A 301.6

B 11.6

C 278.4

D 5.8

Answer: E

Question 39

A differential amplifier is used in the input stage of Op-Ampsto achieve very high

A open —loop gain

B bandwidth

C slew rate

D CMRR

Answer: E

Question 40

Which two-port parameters are best suited for analyzing a series-shunt feedback circuit?

A Z-Parameters

B H-Parameters

C Y-Parameters

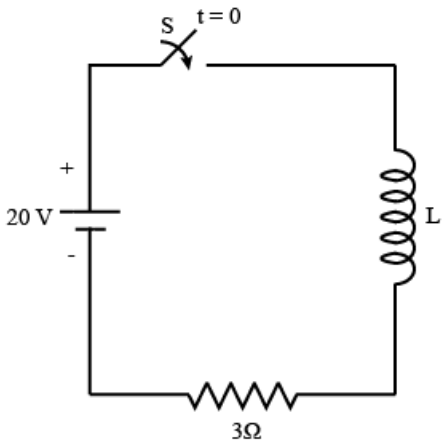
D S-Parameters

Answer: E

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Question 41

For the circuit shown in the figure, the switch is closed at t = 0. After some time when the current in the inductor was 6A,the rate of change of current through the inductor was 4 A/sec. The value of the inductor is



A 0.5 Hf

B 1.5H

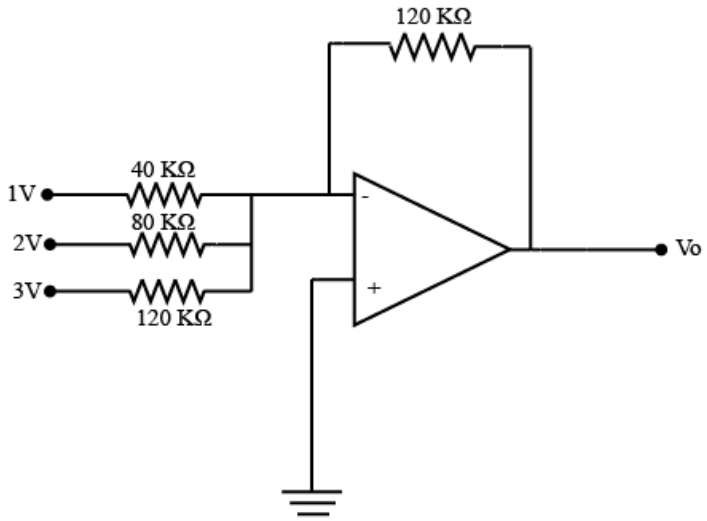
C 1.0H

D Indeterminate

Answer: E

Question 42

In the circuit shown in the above figure, the value of output V_0 is



A +6V

B -9V

C -6V

D +9V

Answer: E

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Question 43

The characteristics equation of a closed loop control system is given as $s^2 + 4s + 16 = 0$. The resonant frequency in radian/sec of the system is

A 2

B $2\sqrt{3}$

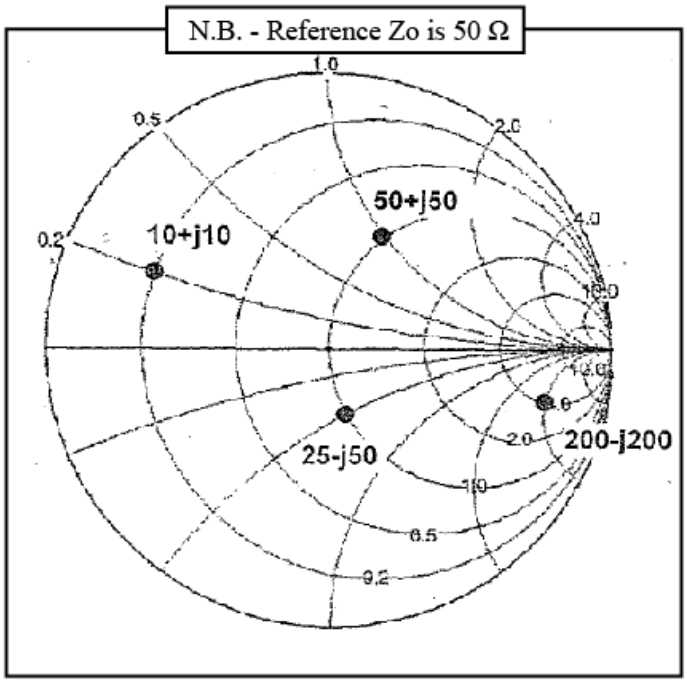
C 4

D $2\sqrt{2}$

Answer: E

Question 44

Which point in the Smith Chart is labeled with the WRONG impedance?



A 10 + j10

B 50 + j50

C 200 - j200

D 25 - j50

Answer: E

Question 45

If a line is terminated in an opencircuit, the VSWRis

A ∞

B 0

C 1

D -1

Answer: E

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Question 46

Unitof relative permeability is

A $\frac{Henry}{meter}$

B Henry

C Henry-meter

D It is dimensionless

Answer: E

Question 47

$f(x) = [\tan^2 x]$

([ ] stands for greatest integer function)

A f(x) continuous at x = 0

B limit f(x) does not exist as x tend to 0

C f'(0)=1

D f(x) not derivable at x = 0

Answer: E

Question 48

Number of real values of $(a+ib)^{\frac{1}{n}} + (a-ib)^{\frac{1}{n}}$

A 0

B 1

C n

D None of the above

Answer: E

Question 49

A 5 bit DAC has a current output. For a digital input of 10100, an output current of 10 mA is produced. What will be the output current for a digital input of 11101?

A 14.5 mA

B = 10mA

C = 100 mA

D Not possible to calculate

Answer: E

Question 50

The ethernet protocol uses

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A $CSMA$ CA

B SCPC

C $CSMA$ CD

D Slotted ALOHA

Answer: E

Question 51

A micro controller differs from a microprocessor in terms of

A I O interfaces and instruction decoding

B Memory configuration and I O interfaces

C Data bus width and clock speed

D Memory configuration and instruction decoding

Answer: E

Question 52

An 8-bit microcontroller has an external RAM with the memory map from 8000H to 9FFFH. The number of bytes this RAM can storeis

A 8193

B 8192

C 8191

D 8000

Answer: E

Question 53

An audio frequency of 15 KHz is frequency modulated with a deviation of 75 KHz. The resulting bandwidthis

A 180 KHz

B 150 KH

C 210 KHz

D 240 KHz

Answer: E

Question 54

A 1kW carrier is Amplitude Modulated to a depth of 60%. The total power in the modulated carrier is

A 1 kW

B 1.06 kW

C 1.6 kW

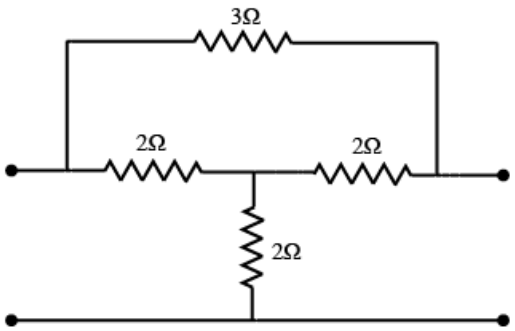
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D 1.18 kW

Answer: E

Question 55

The y_{21} parameter of the network shown in the given figure will be



A $\frac{1}{6}$

B $-\frac{1}{6}$

C $\frac{1}{3}$

D $-\frac{1}{3}$

Answer: E

Question 56

An inductance of 1 His realized using air core with 100 turns. Whatwill be the inductanceif the numberof turns are doubled ?

A 1H

B 0.5H

C 2H

D None of these

Answer: E

Question 57

An inductor of inductance 0.1 H, carrying current of 6 Amps will store energy of

A 6 Joules

B 36 Joules

C 1.8 Joules

D 3.6 Joules

Answer: E

Question 58

A coil has self resistance of SΩ and inductance of 1 μH. The value of Q at 1 MHz frequency is

A 200

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B 100

C 40

D None of these

Answer: E

Question 59

The closed loop transfer function of a unity feedback control system is $G(s) = \frac{1}{(s+2)^2}$.The closed loop transfer function will have poles at

A $-2, \pm j$

B $-2, -1$

C $-2, -2$

D $-2, 2$

Answer: E

Question 60

The dominant modeof propagation in microstrip line in low frequency approximation is

A TM_{11}

B Quasi-TEM

C TEM

D TE_{01}

Answer: E

Question 61

Consider a transmission line of characteristic impedance 50 ohms and theline is terminated at one end by +j50 ohms, the VSWRproducedin the transmission line will be

A +1

B zero

C -1

D infinity

Answer: E

Question 62

The divergence of magnetic flux density is

A j

B ρ

C 0

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D ∞

Answer: E

Question 63

A straight line and a circle of radius a are given. A chord is drawn at random to this circle parallel to the given line. Expected length of the chordis

A 2a

B πa

C $\frac{\pi a}{2}$

D None of the above

Answer: E

Question 64

There is a function $f(x)$, such that $f(0) = 1$ and $f'(0) = -1$ and $f(x)$ is positive for all values of x. Then,

A $f"(x) < 0$ for all x

B $-1 < f"(x) < 0$ for all x

C $-2 < f"(x) < -1$ for all x

D None of the Above

Answer: E

Question 65

Interrupt latencyis the time elapsed between:

A Occurrence of an interrupt and its detection by.the CPU

B Assertion of an interrupt and the start of the associated ISR

C Assertion of an interrupt and the completion of the associated ISR

D Start and completion of associated ISR

Answer: E

Question 66

Inverse Fourier transform of signal
$X(j\omega) = 2\pi\delta(\omega) + \pi\delta(\omega - 4\pi) + \pi\delta(\omega + 4\pi)$

A $1 + \cos 4\pi t$

B $\pi(1 - \cos 4\pi t)$

C $2\pi(1 - \cos 4\pi t)$

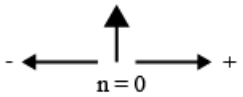
D $2\pi(1 + \cos 4\pi t)$

Answer: E

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Question 67

Determine the discrete-time Fourier Transform for the signal $x[n] = [-2, -1, 0, 1, 2]$



A $2j(2\sin 2\omega + \sin\omega)$

B $-2j(2\sin 2\omega + \sin\omega)$

C $2(2\cos 2\omega - \cos\omega)$

D $-2(2\cos 2\omega - \cos\omega)$

Answer: E

Question 68

Auto-correlation of a sinusoid $s(t) = A\sin(\omega t)$ **is**

A $(\frac{A^2}{2})\sin(\omega\tau)$

B $(\frac{A}{2})\sin(\omega\tau)$

C $(\frac{A^2}{2})\cos(\omega\tau)$

D $(\frac{A}{2})\cos(\omega\tau)$

Answer: E

Question 69

A source produces 4 symbols with probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}$ **For this source, a practical coding scheme has an average codeword length of 2 bits/symbol. The efficiency of the code is**

A 1

B $\frac{7}{8}$

C $\frac{1}{2}$

D $\frac{1}{4}$

Answer: E

Question 70

When signal frequency is 2000 KHz and IF is 455 KHz,the image frequency could be

A 2910 KHz

B 1645 KHz

C 2455 KHz

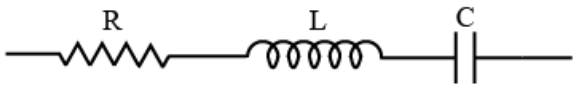
D 2090 KHz

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Answer: E

Question 71

The Q of RLC tuned circuit with source voltage V in the figure is 100 at resonant freuquency of 100 KHZ What is the voltage V_L across inductor at resonance ?



A 50V

B $V_L - V_C$

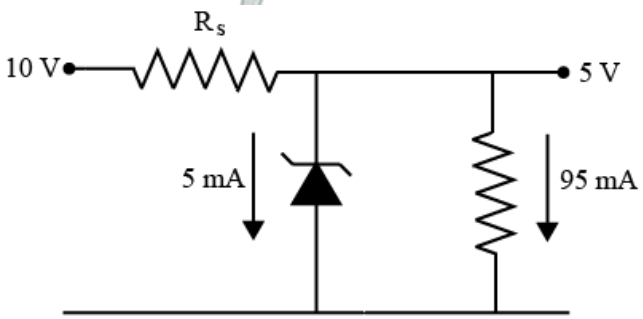
C 100V

D $V_R - V_C$

Answer: E

Question 72

A 5 V reference is drawn from the circuit shown in the figure. Zener diode of 400 mW and 5V with firing current of 5 mA is used. The value of R_s is



A 50 ohms

B 500 ohms

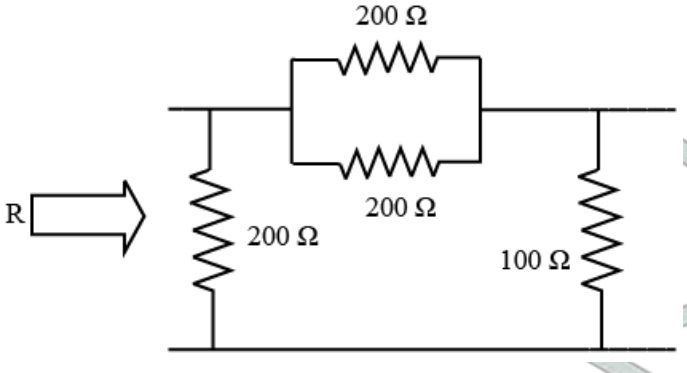
C 75 ohms

D 470 ohms

Answer: E

Question 73

Find the equivalent resistance 200 Q R for the network shown in the figure.



A 200 ohms

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B 101 ohms

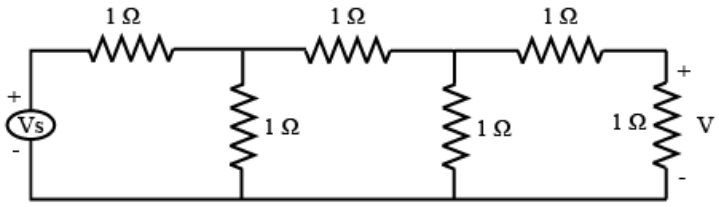
C 220 ohms

D 100 ohms

Answer: E

Question 74

The Voltage across last resistor is V. All resistors are unity. Then V_s is given by



A 13V

B 8V

C 4v

D None of these

Answer: E

Question 75

Natural frequency of a unity feedback control system of transfer function G(s)= $\frac{10}{s(s+1)}$

A 0.5 rad/sec

B 3.16 rad/sec

C 4.6 rad/sec

D None of these

Answer: E

Question 76

Which ofthe following conditions will not guarantee a distortionless transmission line?

A R=0, G=0

B Very low frequency range $(R >> \omega L, G >> \omega C)$

C $\frac{R}{L} = \frac{G}{C}$

D Very high frequency range $(R << \omega L, G << \omega C)$

Answer: E

Question 77

For a line of characteristic impedance Z_o terminated in a load Z_R such that $Z_R > Z_0$, the Voltage Standing Wave Ratio (VSWR)is given by

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A $\frac{Z_R}{Z_0}$

B Z_0

C Z_R

D $\frac{Z_0}{Z_R}$

Answer: E

Question 78

The beam width of a reflector antenna of diameter = 70 cm at 30 GHz is

A 3.7 degree

B 0.5 degree

C 1.1 degree

D 1 degree

Answer: E

Question 79

$T_r = \overset{r}{1+r^2+r^4}$

$\sum_1^\infty T_r =$

A $\frac{1}{4}$

B 1

C None of the aove

D $\frac{1}{2}$

Answer: E

Question 80

A man with n keys wants to open a lock. He tries his keys at random. The expected number of attempts for his success is (keys are replaced after every attempt)

A $\frac{n}{2}$

B n

C $\sqrt{n}$

D None of the above

Answer: E

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